EXPERIMENT NO. : 4.1

CUT SECTION OF SOLIDS

AIM

To model the given 3D component (Flange Bracket / Bearing Mount) using Autodesk Fusion 360 software based on the provided dimensional engineering drawing.

REQUIREMENTS

Hardware Requirements:

- Processor (CPU): 64-bit x86-based processor
- Memory (RAM): Minimum 8 GB (16 GB or higher recommended)
- Graphics Card (GPU): Dedicated GPU with 1 GB VRAM minimum
- Disk Space: Approximately 8.5 GB of free storage space
- Input Devices: 3-button mouse with scroll wheel

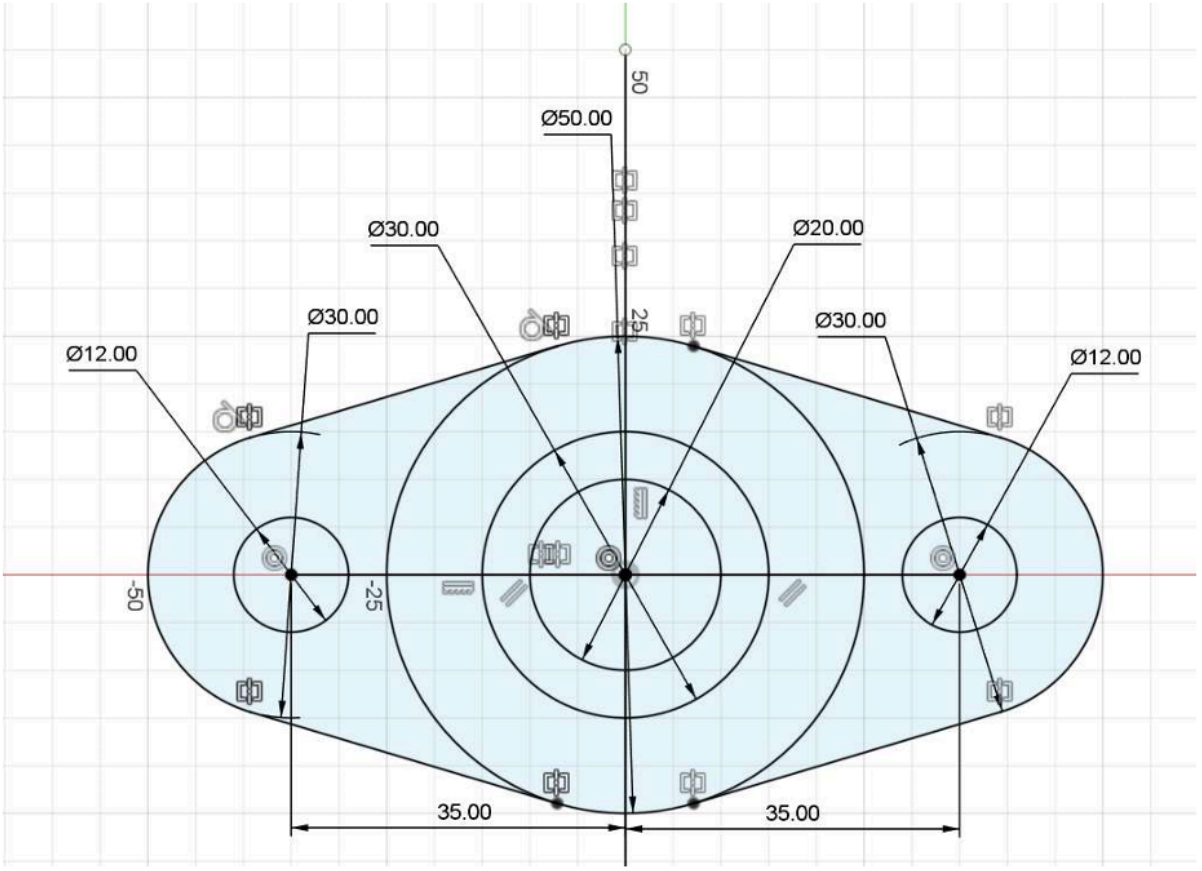
Software Requirements:

- Operating System: Microsoft Windows 11 / 10 (64-bit) or macOS (12.0 Monterey or later)
- CAD Software: Autodesk Fusion 360 (Active License / Student Edition)

PROCEDURE

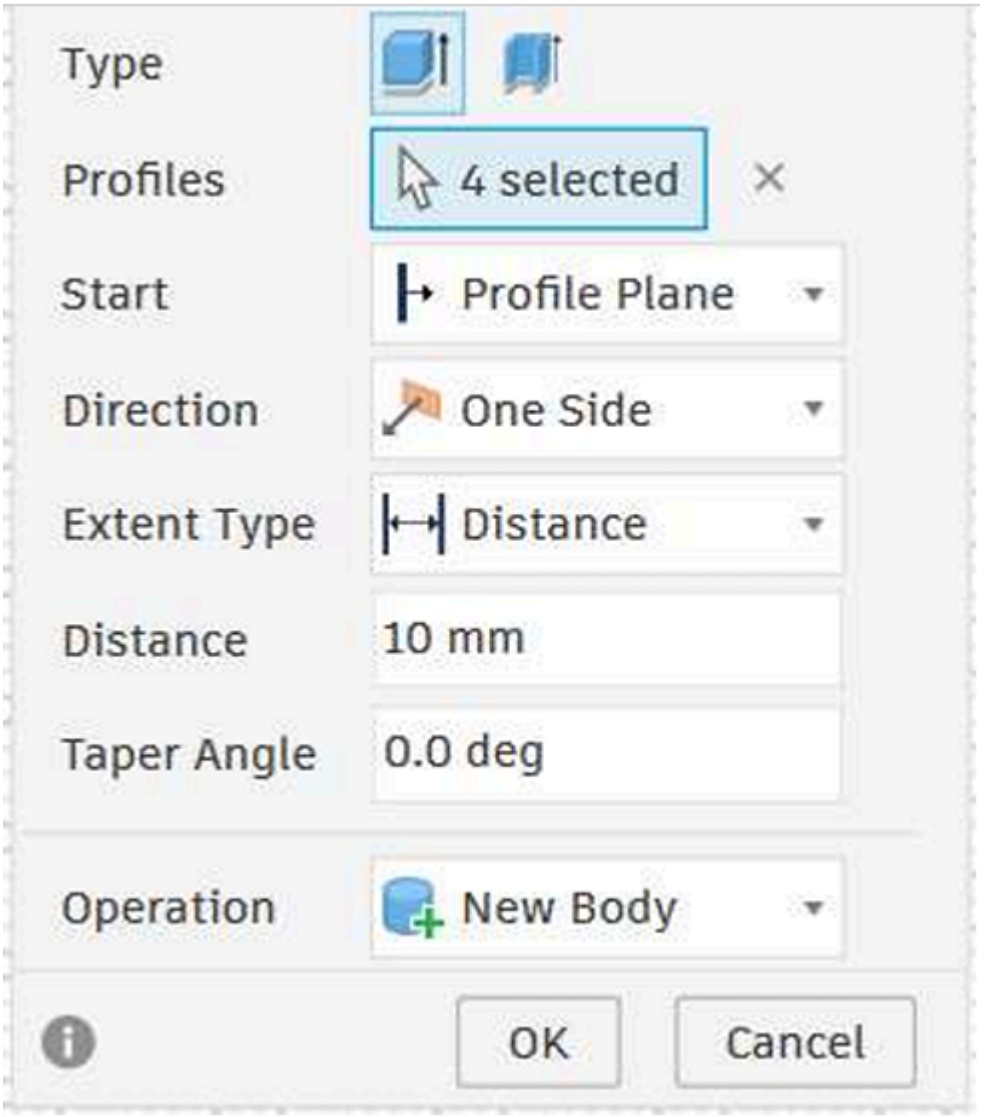
Step 1: Create the Base Flange Profile

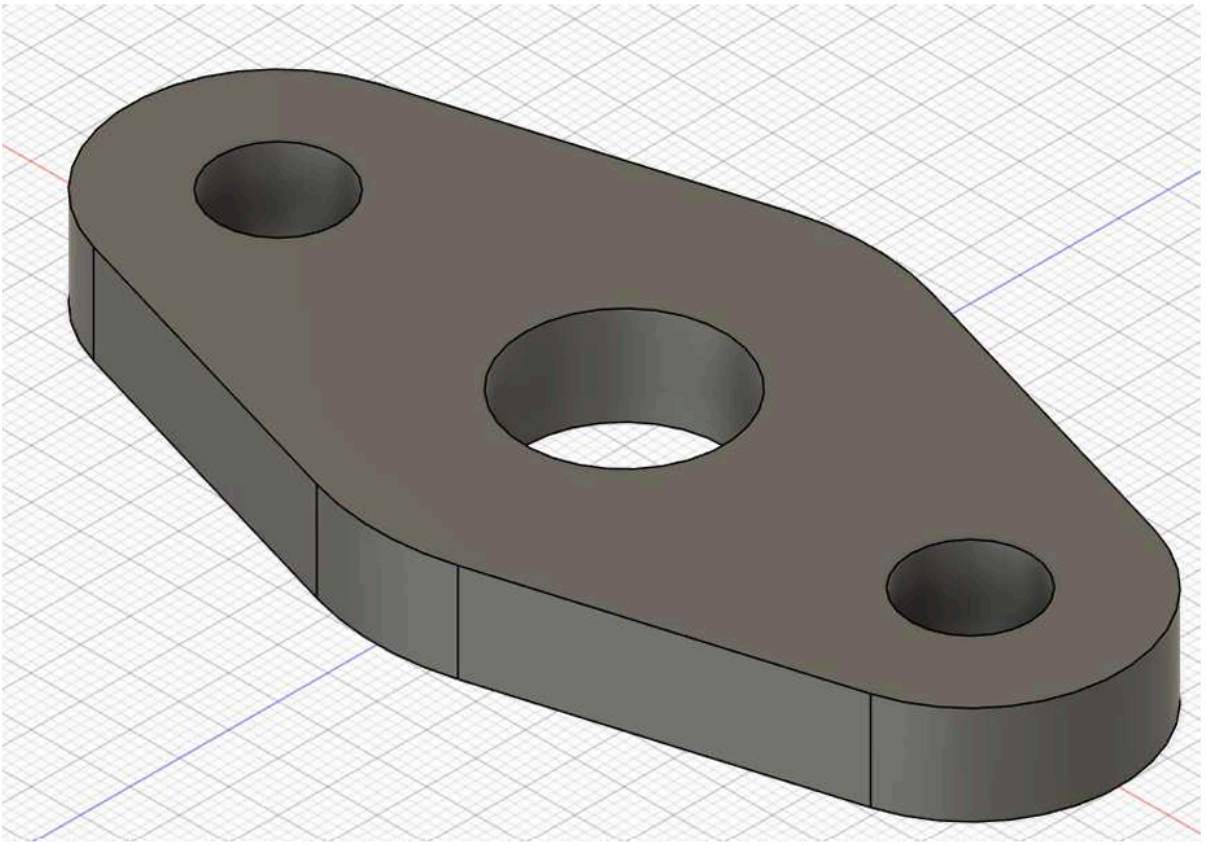
1. Click on Create Sketch and select the XY Plane (Top Plane).
2. Draw the main centerline along the X-axis.
3. Draw a central circle of Diameter 50mm centered at the origin (0, 0).
4. Draw two circles of R15mm (Diameter 30 mm) centered at (-35,0) and (35,0) relative to the origin.
5. Draw two mounting hole circles of diameter 12 mm centered at the same points (-35,0) and (35,0).
6. Draw smooth tangent lines connecting the outer arc circles (R15) to the central cylinder profile (diameter 50 mm) on both top and bottom sides using the Line tool and applying Tangent Constraints.
7. Trim any unnecessary inner geometry using the Trim tool (T) to create a clean, closed outer boundary outline for the flange base.
8. Click Finish Sketch.



Step 2: Extrude the Base Flange

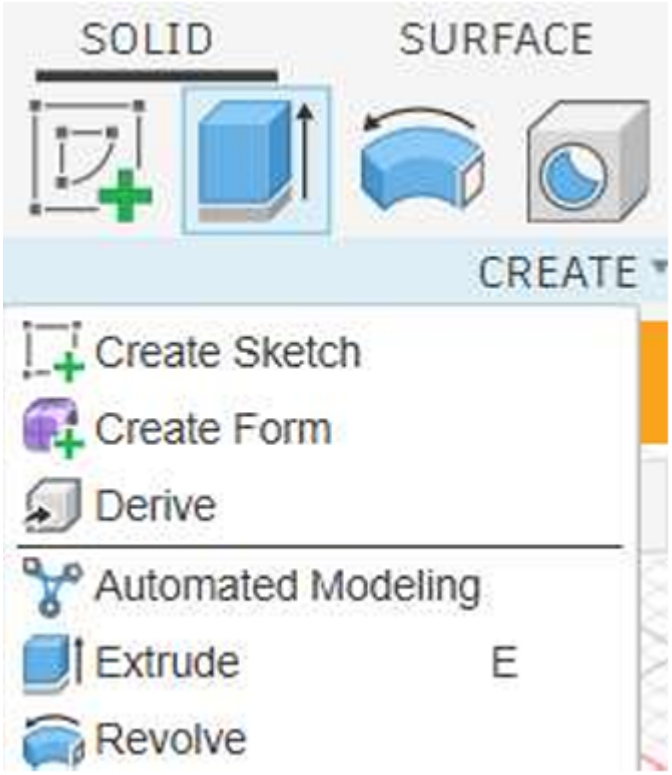
1. Select the Extrude tool (E).
2. Select the base flange profile (excluding the two diameter 12 mm holes).
3. Enter Distance = 10mm in the upward direction.
4. Set Operation to New Body and click OK.


Type
Profiles
4 selected
Start
Profile Plane
Direction
One Side
Extent Type
Distance
Distance
10 mm
Taper Angle
0.0 deg
Operation
New Body
OK
Cancel

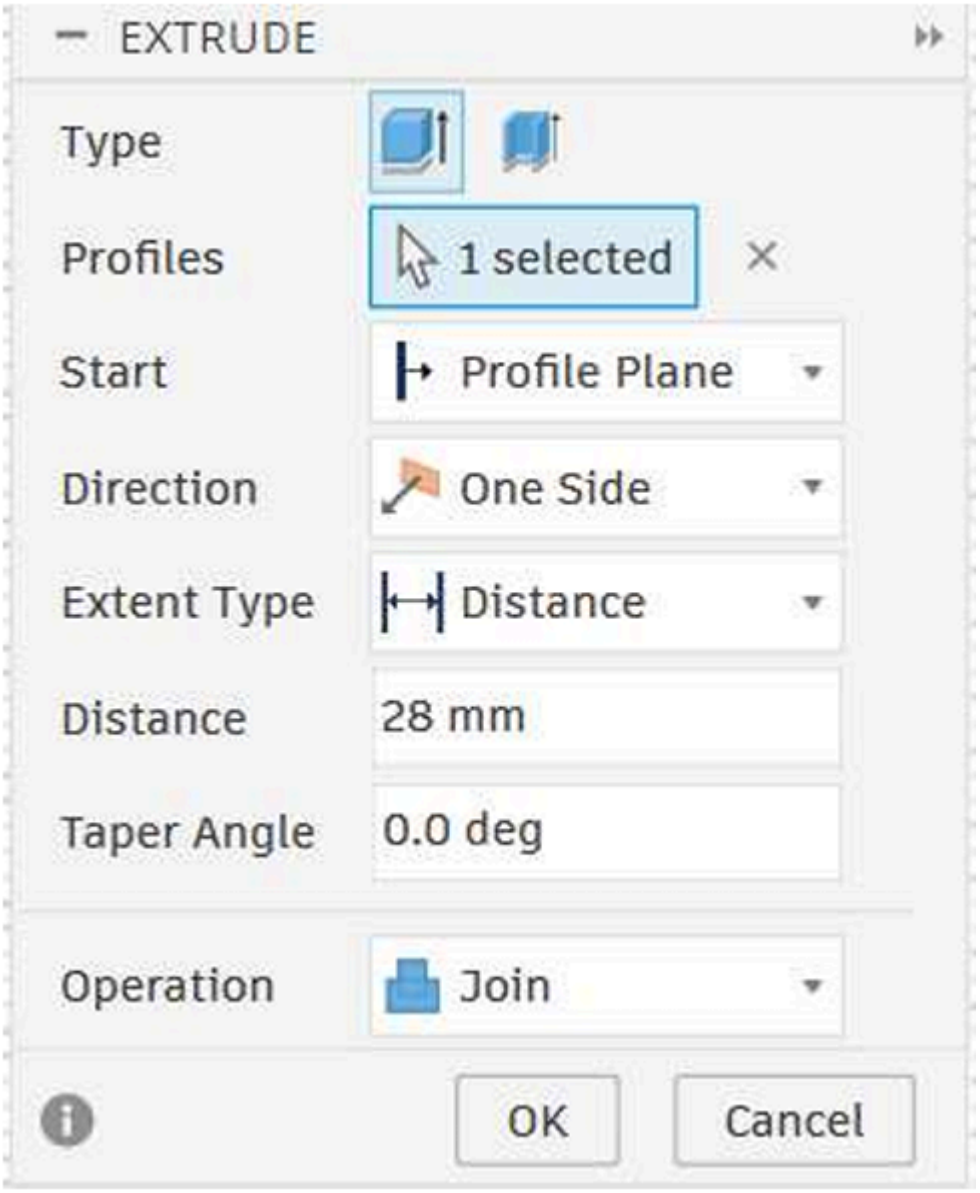


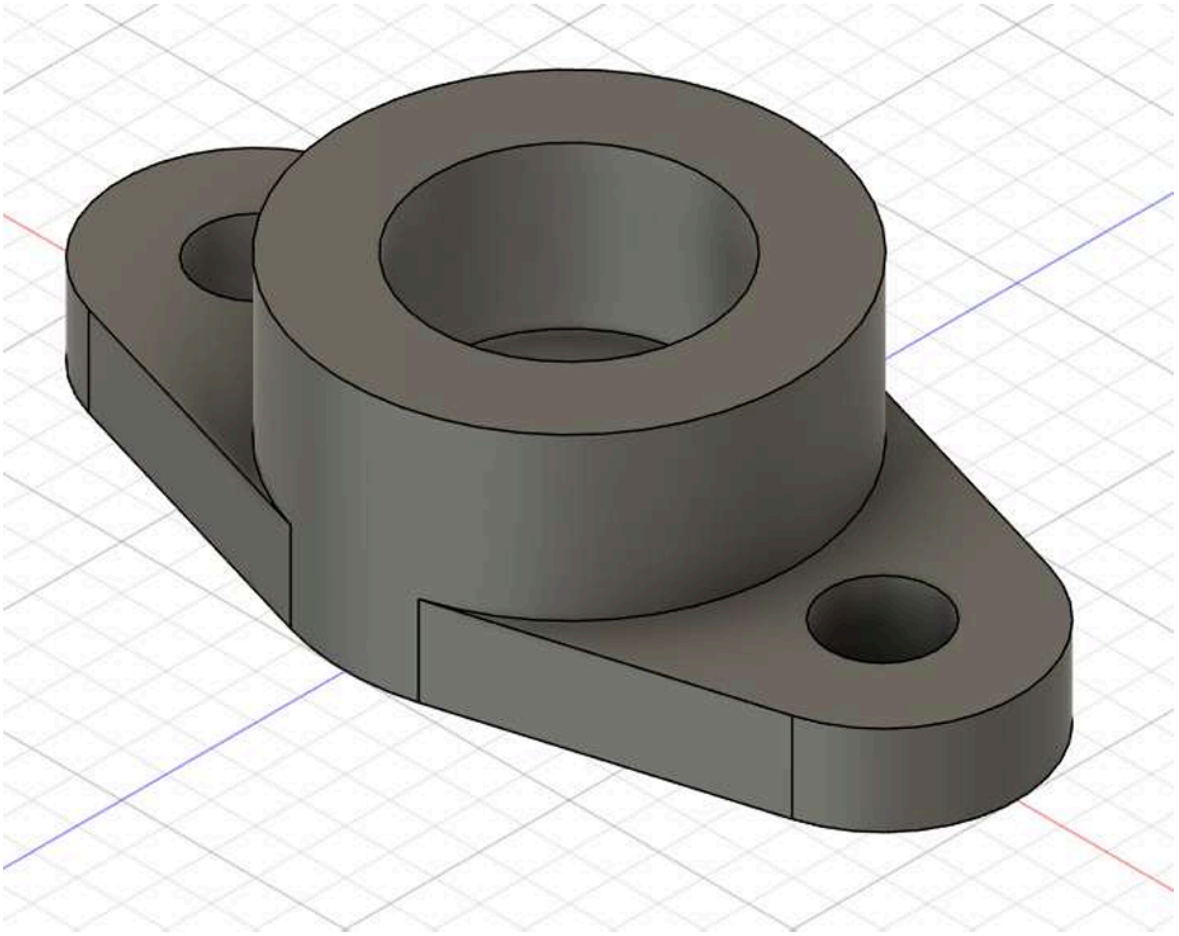
Step 3: Create and Extrude the Central Boss Cylinder

1. Click Create Sketch and select the top flat face of the newly created base flange (or the XY Plane).
2. Draw a circle centered at the origin with Diameter = 50 mm.
3. Click Finish Sketch.
4. Select the Extrude tool (E).



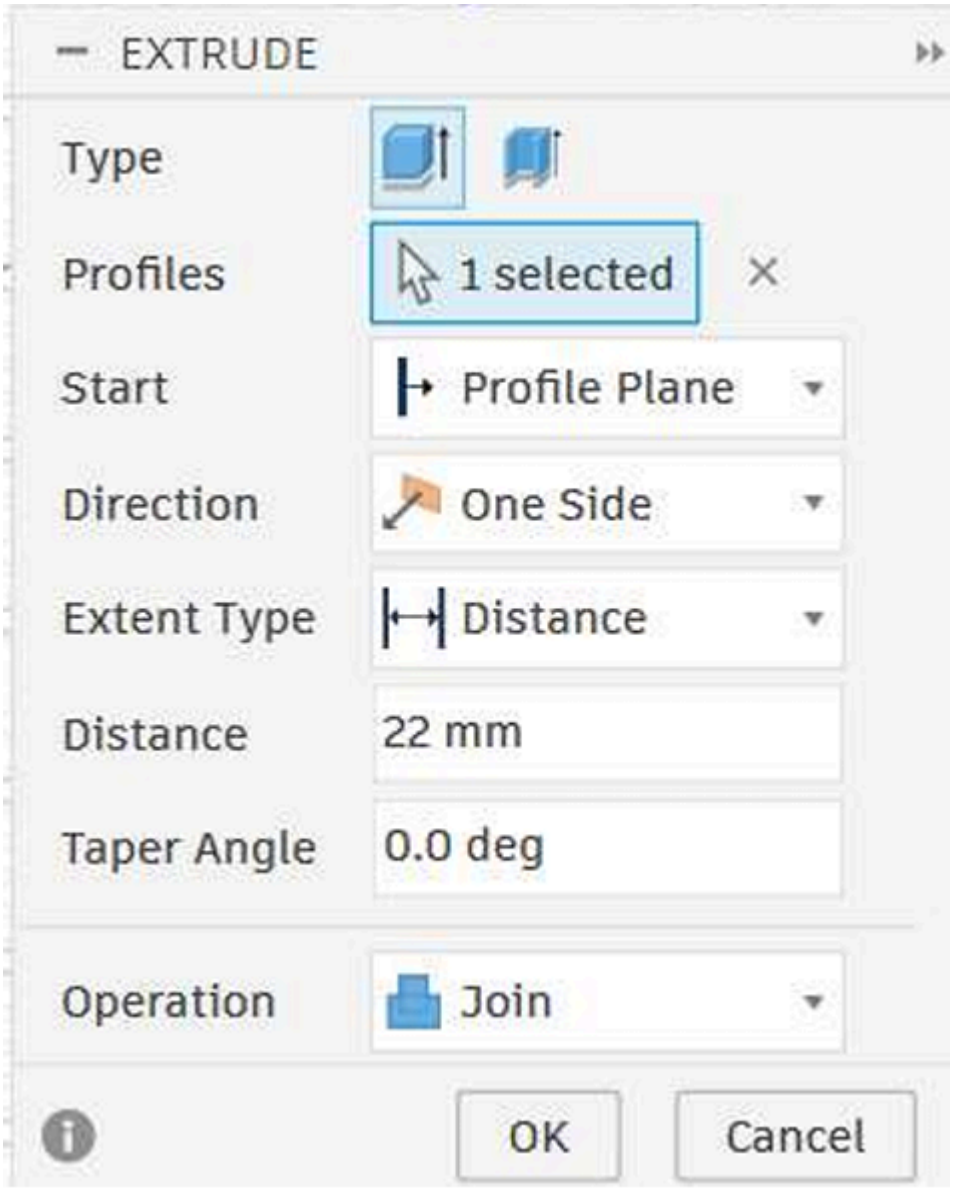
5. Select the 50 mm circular profile.
6. Set Distance = 18mm (giving a total height of 28mm from the bottom face: 10mm + 18mm = 28mm).
7. Set Operation to Join and click OK.

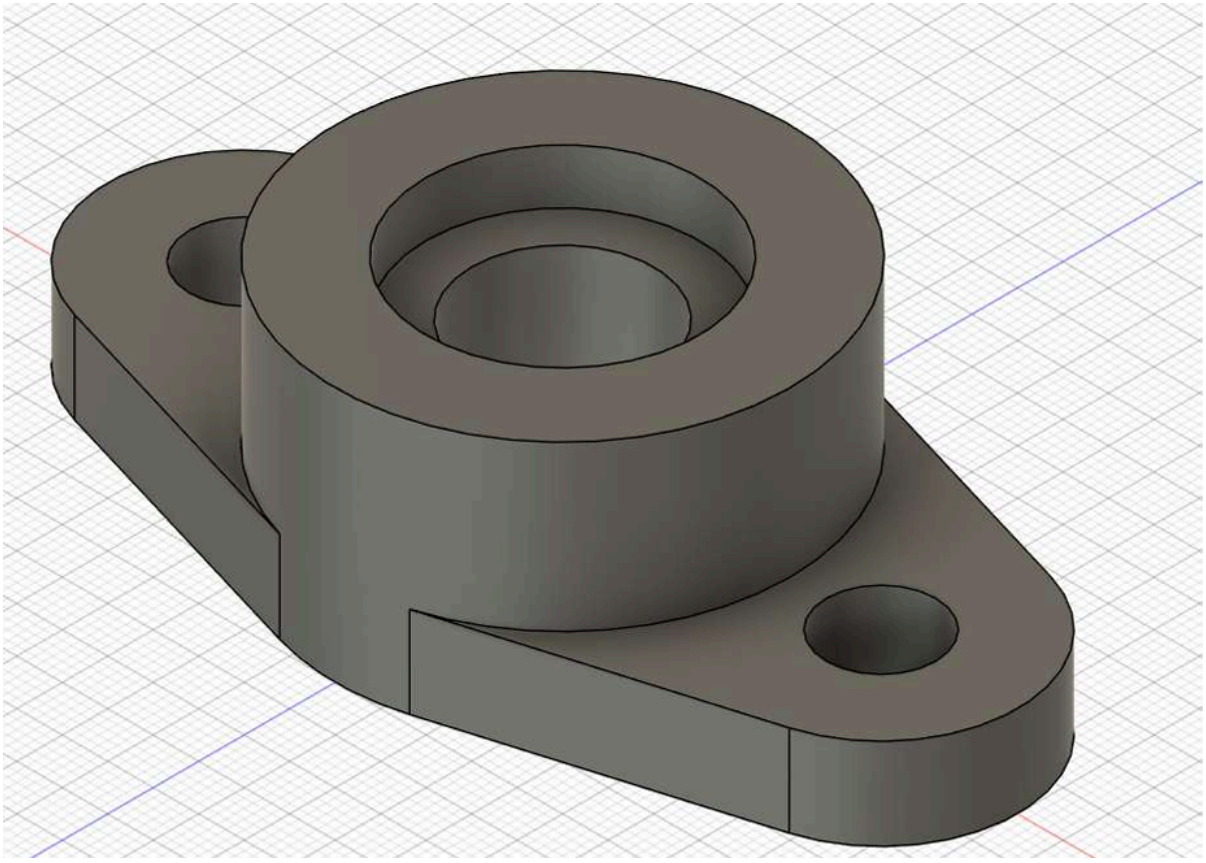

EXTRUDE
Type
Profiles
1 selected
Start
Profile Plane
Direction
One Side
Extent Type
Distance
Distance
28 mm
Taper Angle
0.0 deg
Operation
Join
OK
Cancel



Step 5: Create the Stepped Counterbore Hole

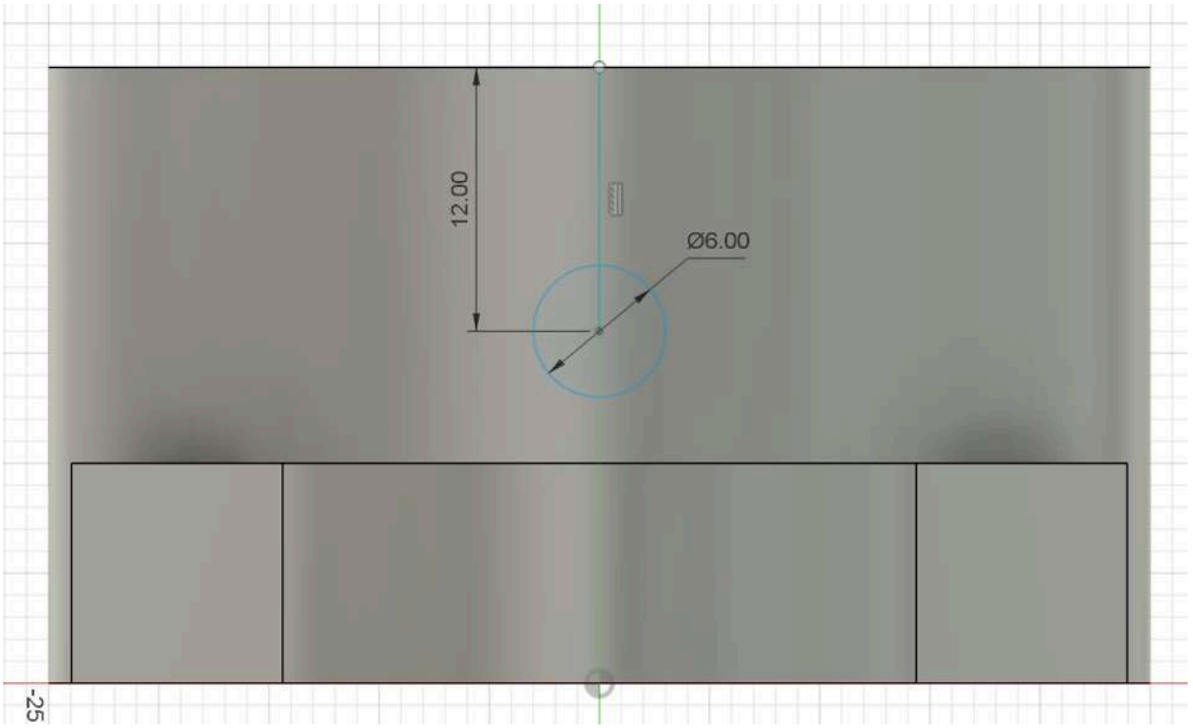
Select the Extrude tool again, select the inner annular region profile (between diameter 30 mm and diameter 20 mm), set the distance upward to 22mm (28mm total height minus the 6mm counterbore depth), set the operation to Join, and click OK.


EXTRUDE
Type
Profiles
1 selected
Start
Profile Plane
Direction
One Side
Extent Type
Distance
Distance
22 mm
Taper Angle
0.0 deg
Operation
Join
OK
Cancel

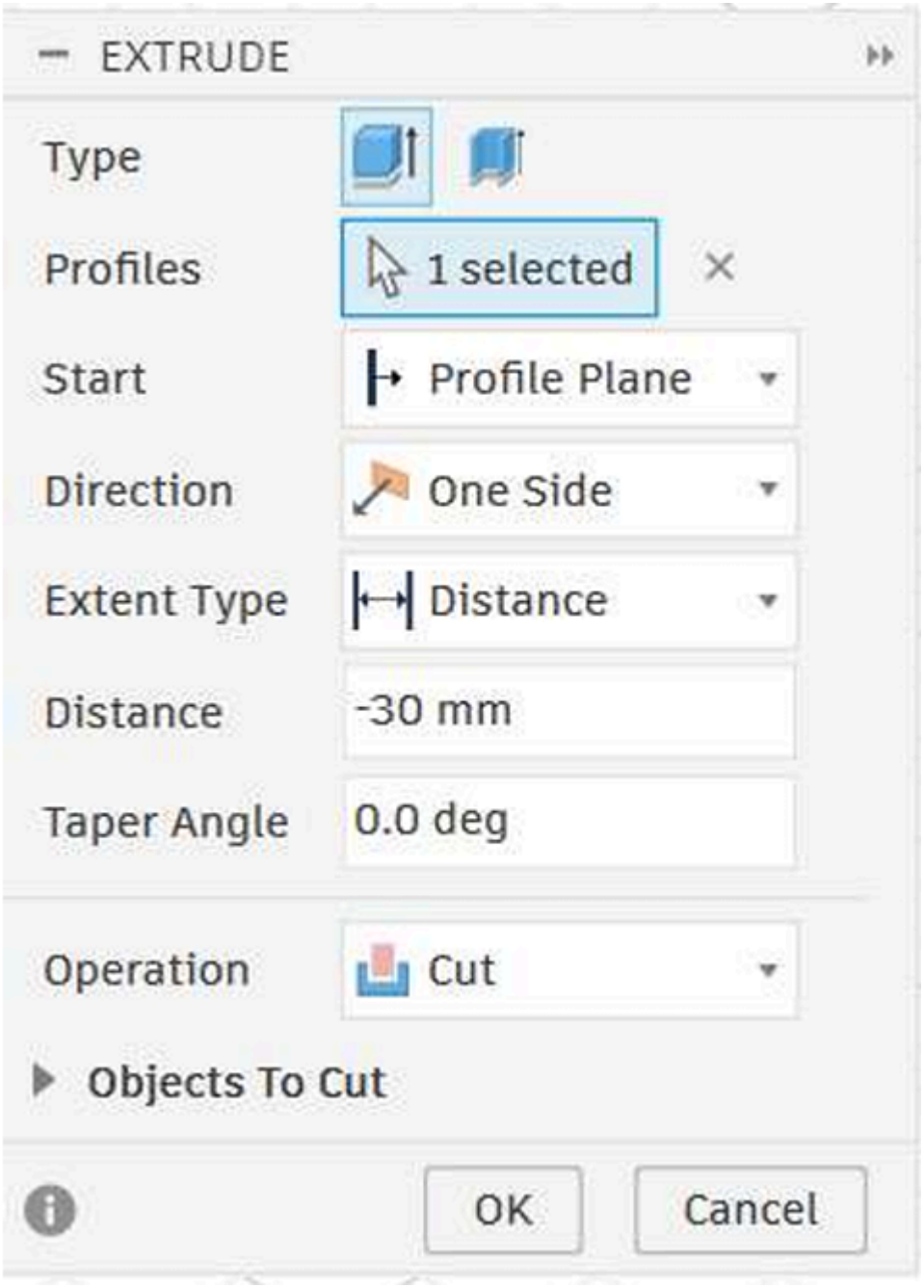


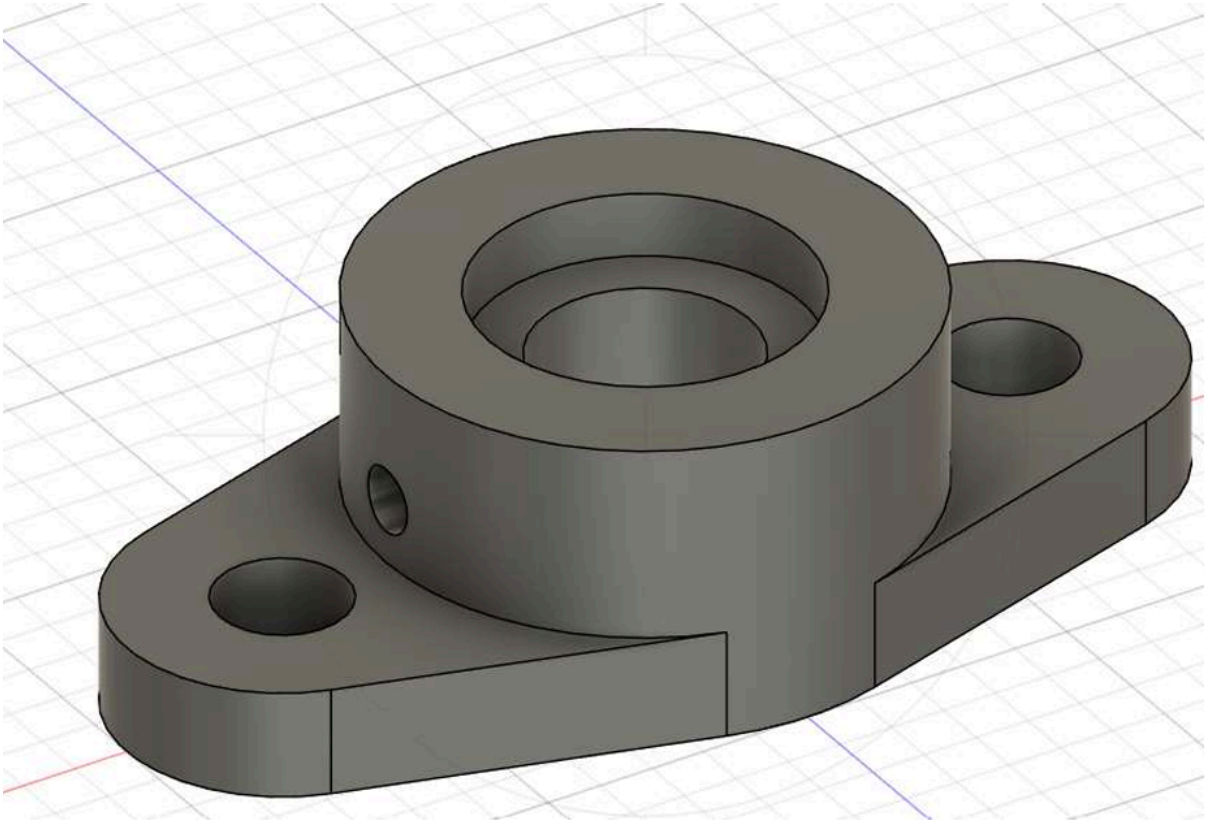
Step 6: Create the Side Cross Hole (Diameter = 6 mm)

1. Construct an offset plane or select an orthogonal construction plane (XZ Plane or YZ Plane) passing through the center of the cylinder.
2. Click Create Sketch on that plane.
3. Draw a circle with Diameter = 6 mm.
4. Dimension its position:
    - Vertical distance: 12mm down from the top face (or 16mm up from the bottom base line).
    - Align the center vertically with the central axis.



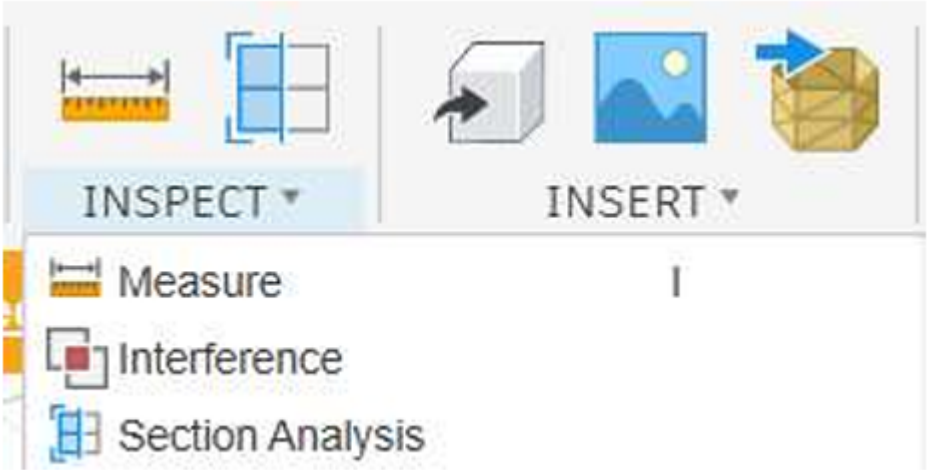
5. Click Finish Sketch.
6. Select the Extrude Tool (E).
7. Select the diameter = 6 mm circle profile.
8. Set Direction to One Side (extruding towards the inner cavity) or Symmetric/Through Wall.
9. Set Operation to Cut and click OK.


EXTRUDE
Type
Profiles
1 selected
Start
Profile Plane
Direction
One Side
Extent Type
Distance
Distance
-30 mm
Taper Angle
0.0 deg
Operation
Cut
Objects To Cut
OK
Cancel

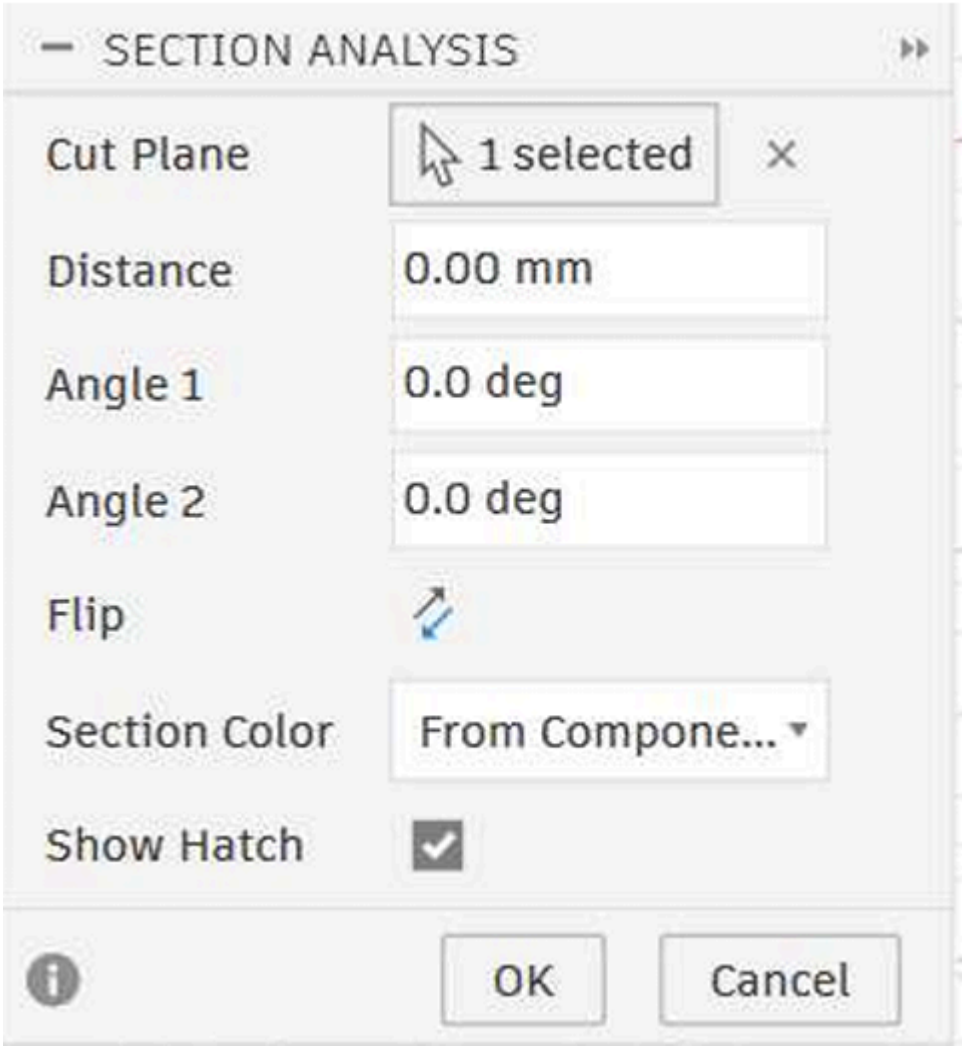


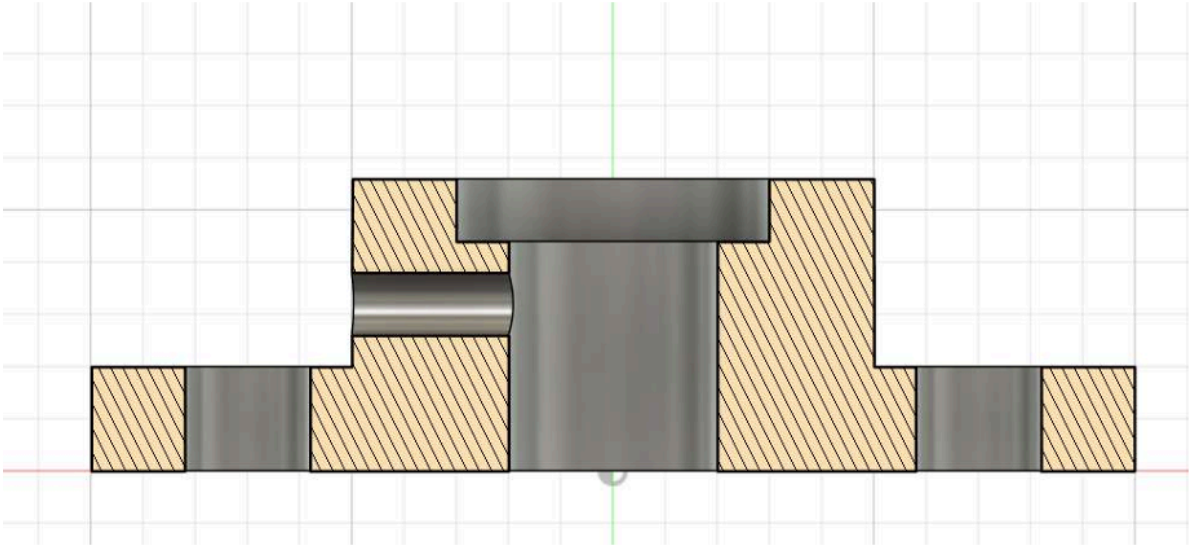
Step 7: Inspect Internal Features via Section Analysis

1. Go to the top application menu bar, click on Inspect, and select Section Analysis.
2. Choose an orthogonal construction plane (such as the XZ or YZ plane) passing through the center origin of the model.

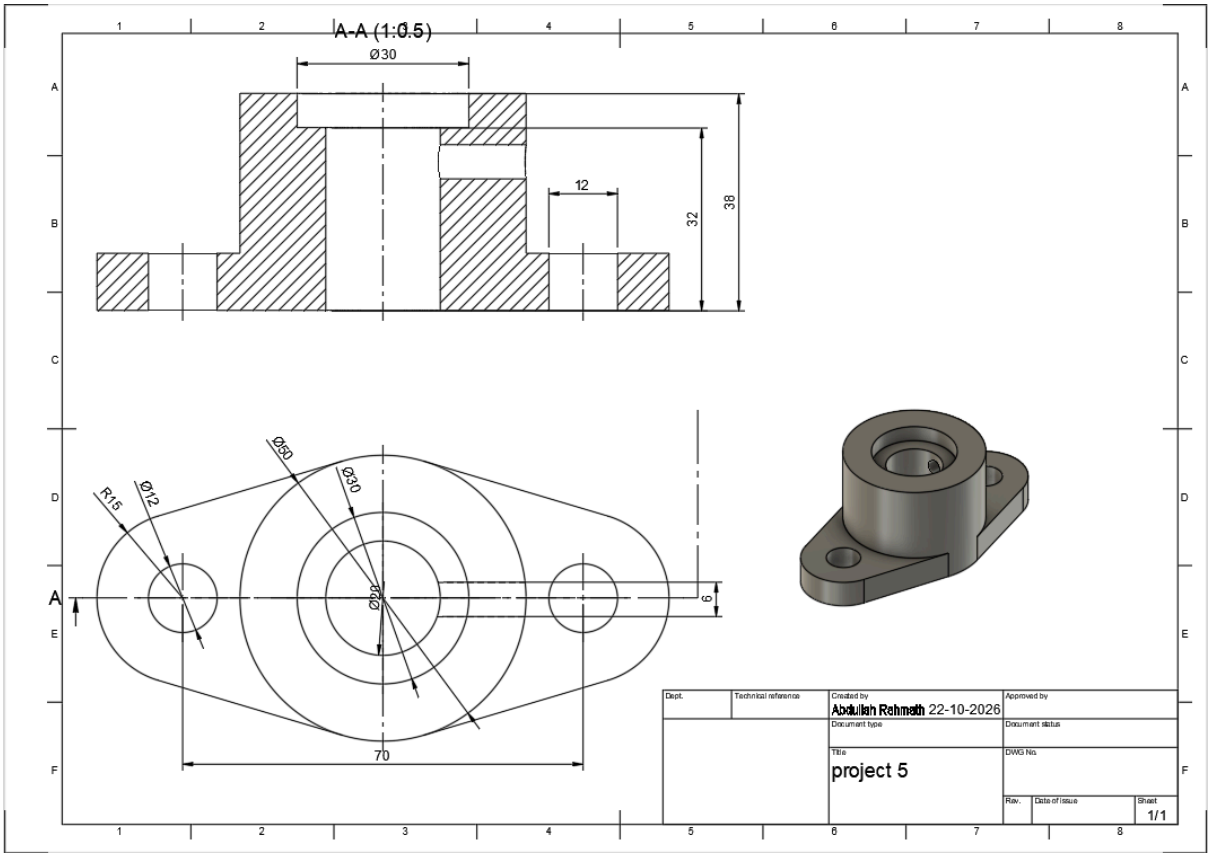


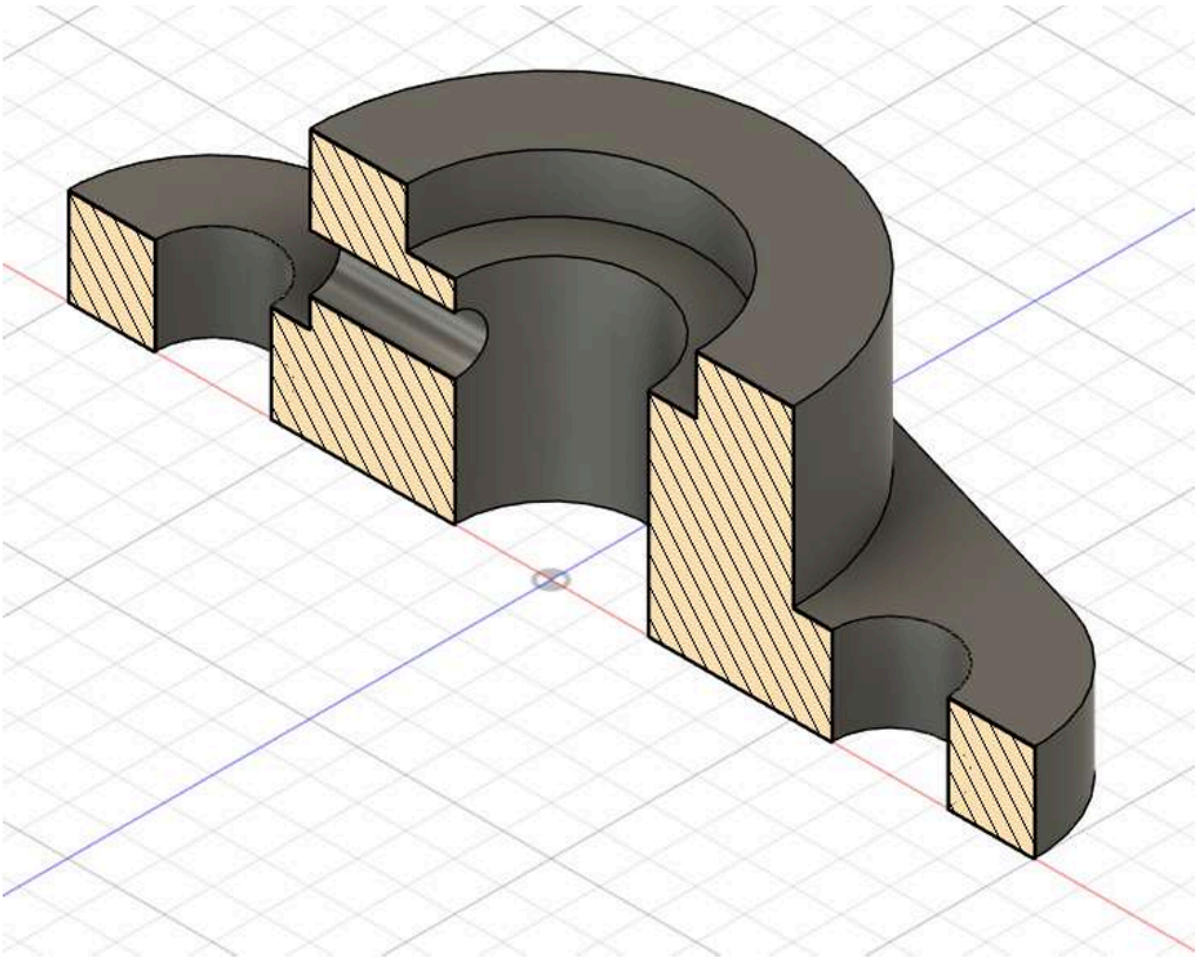
3. Set the offset distance to 0mm to cut the object in half vertically along the centerline, and click OK to dynamically inspect internal features like the stepped counterbore, through-hole, and side lubrication hole.


SECTION ANALYSIS
Cut Plane
1 selected
Distance
0.00 mm
Angle 1
0.0 deg
Angle 2
0.0 deg
Flip
Section Color
From Compone...
Show Hatch
OK
Cancel



OUTPUT:


A-A (1:0.5)
Ø30
12
32
38
Ø50
Ø30
R15
Ø12
Ø20
6
70
Dept.
Technical reference
Created by
Abdullah Rahmath 22-10-2026
Approved by
Document type
Document status
Title
project 5
DWG No.
Rev.
Date of issue
Sheet
1/1



RESULT:

Thus, the cut section for solid model is created.